

Lorenzo Liuzzo

PHYSICIST · DEVELOPER · SOFTWARE ARCHITECT

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Physicist and developer specialized in AI, combining *machine learning* expertise with robust *software engineering* to solve complex problems through scientific rigor and creative innovation.

Education

University of Milan, Milan-Bicocca and Pavia

MASTER'S DEGREE IN ARTIFICIAL INTELLIGENCE FOR SCIENCE AND TECHNOLOGY

Milan/Pavia, Italy

September 2025 - Present

University of Milan

BACHELOR'S DEGREE IN PHYSICS, GRADE: 100/110

Milan, Italy

September 2021 - July 2025

Ray tracing and Monte Carlo methods for uncertainty quantification in the pointing direction of radio telescopes for CMB observations.

I.I.S. Severi-Correnti

SCIENTIFIC DIPLOMA, GRADE: 100/100

Milan, Italy

September 2016 - June 2021

EF International Language School

FULL-IMMERSION INTENSIVE ENGLISH LANGUAGE COURSE

Brighton, England

June - September 2019

Schools & Workshops

School on Quantum Simulation

DEPARTMENT OF PHYSICS, UNIVERSITY OF MILAN

Milan, Italy

9 - 11 September 2026

Qiskit Fall Fest

IBM QUANTUM

Milan, Italy

Autumn 2025

Experiences

Money Ball AI

FOUNDER, TECHNICAL LEAD AND DEVELOPER

Milan, Italy

September 2025 - Present

- Founded and lead *MoneyBallingAI*, a team of four (physics MSc student, CS MSc graduate, statistician) turning raw NBA play-by-play data into predictive models for basketball analytics.
- Own the platform architecture: a graph-database backend for play-by-play data, a simulation backend, and the Machine Learning layer built on top of them.

Private Tutoring in Mathematics and Physics

IN-HOME TUTORING

Milan, Italy

2020 - Present

- Continuous support for 20+ students
- High school graduation exam preparation and remedial tutoring
- Personalized teaching methodologies tailored to student needs
- Progress monitoring and periodic reports to families

Barista

DON SALVATORE - PIZZAIUOLO E OSTE

Milan, Italy

September - October 2024

Waiter

BAR TERRAZZA CLÉR

Milan, Italy

May - July 2022

Scout Leader

REPARTO BREITHORN, GRUPPO MILAN 8, AGESCI

Milan, Italy

2021-2022

- Design and leadership of educational and recreational activities for youth aged 12-16
- Organization of winter/summer camps and educational outings throughout the year
- Coordination with other educators and families

Projects

MyThingsLab

 MyThingsLab

ARCHITECT/DEVELOPER

July 2026 - Present

- A fleet of 56 composable Python tools on a shared SDK, with a policy engine gating every side effect as allow/ask/deny.
- Drives an autonomous development loop: headless LLM workers pick issues off a backlog and close them as pull requests, deterministic except where judgement is genuinely needed.

Gauge-Equivariant Graph Neural Networks

 lorenzoliuzzo/EGNN

DESIGNER/DEVELOPER

July - August 2026

Lattice Standard Model field theory simulated with gauge-equivariant GNNs in *PyTorch Geometric*, covering $SU(3) \times SU(2) \times U(1)$ gauge groups, Wilson-Dirac fermions and Higgs spontaneous symmetry breaking.

Supervised Learning on Food Images

 lorenzoliuzzo/supervised-learning-on-food-images

DESIGNER/DEVELOPER

July 2026

A convolutional neural network under 10M parameters classifying the 251 classes of the FoodX-251 dataset.

Transverse Field Ising Model

 lorenzoliuzzo/TFIM

DESIGNER/DEVELOPER

July 2026

Exploration of quantum phase transitions in the Transverse Field Ising Model, implemented on *PennyLane*.

Unsupervised Learning on Country Data

 lorenzoliuzzo/unsupervised-learning-on-country-data

DESIGNER/DEVELOPER

May 2026

Clustering and dimensionality reduction applied to socio-economic country-level data.

Telescope Pointing Simulation

 lorenzoliuzzo/thesis-L30

DESIGNER/DEVELOPER

June - July 2025

Implementation in *Julia* that allows modeling the reflective system of an optical telescope and studying the propagation of instrumental defects in mirrors to the pointing direction, combining geometric optics with *ray tracing* and *Monte Carlo* simulation techniques.

Ray Tracing Engine

 lorenzoliuzzo/PhotoNim

CO-DESIGNER/DEVELOPER

March - July 2024

- Pair project for the course *Numerical Techniques for Photorealistic Image Generation* at UNIMI.
- Translated a physical model into well-structured, tested and documented numerical code, while rapidly learning a new language (*Nim*).

Dimensional Analysis Library

 lorenzoliuzzo/ctda-cpp

DESIGNER/DEVELOPER

September - October 2023

“CTDA: Compile Time Dimensional Analysis”, implementation in C++ of numeric types supporting units of measurement.

Skills

Languages

Italian, English C1

Programming Languages

C++, Python, Julia, C, Nim

Markup Languages

LaTeX, Typst, Markdown, HTML

Development Tools

Git, GitHub, Jupyter

ML & Data Science

PyTorch, PyTorch Geometric, Scikit-learn, Pandas, NumPy, SciPy

Quantum Computing

PennyLane, Qiskit

Personal Interests

Hiking & Camping

Walking in the mountains and experiencing the essentiality of nature

Volunteering

Scout camps, Elderly assistance, Music therapy for children with disabilities

Sports

Swimming, Water polo, SlackLine, Climbing

Travel

Any destination and mode: on foot, by bike, by train, by car, by plane